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Restoration Intelligence Assessment Report

Jonnahalli Lake 2 — Restoration Feasibility &
Satellite-Based Ecological Intelligence Assessment

Reference	NEV-RIA-JHL2-2026-002
Version	1.0 — Pre-Feasibility Assessment
Date	May 25, 2025
Location	Jonnahalli Lake 2, Devanahalli, Karnataka
Prepared By	NEVARA Ecological Intelligence Division
Classification	Confidential — Pre-Feasibility Ecological Intelligence

■ CONFIDENTIAL PRE-FEASIBILITY DOCUMENT ■

EXECUTIVE INTELLIGENCE SUMMARY

Satellite-derived ecological indicators and restoration feasibility overview for Jonnahalli Lake 2.

This Restoration Intelligence Assessment presents a satellite-derived ecological analysis of Jonnahalli Lake 2, Devanahalli Region, Karnataka, produced by the NEVARA Ecological Intelligence Division as a pre-feasibility decision-support document. The assessment establishes observational baselines for restoration planning, CSR evaluation, and MRV framework integration using Sentinel-2 MSI and Landsat 8/9 data processed through Google Earth Engine.

VERIFIED SATELLITE OBSERVATIONS

MEAN NDVI 0.238 SPARSE VEGETATION	MEAN NDWI -0.278 LOW MOISTURE	MEAN LST 34.64°C ELEVATED THERMAL	SUITABILITY 0.389 MODERATE POTENTIAL	ECO. STRESS -0.240 MODERATE STRESS
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ASSESSMENT CONTEXT

Water Body	Jonnahalli Lake 2	Region	Devanahalli, Karnataka
Assessment Type	Pre-Feasibility Ecological Intelligence	Satellite Sources	Sentinel-2 MSI, Landsat 8/9
Processing Platform	Google Earth Engine	Assessment Period	May 2026 — Dry Season
Climate	Tropical Monsoon (Aw)	Document Ref	NEV-RIA-JHL2-2026-002

KEY OPERATIONAL INTELLIGENCE

INTELLIGENCE LAYER	ASSESSMENT STATUS	SOURCE
Restoration Priority	Shoreline stabilization and runoff interception — primary Phase 1 interventions	Satellite-derived
Cost-to-Impact	Native grass establishment yields highest ecological return per investment unit	Interpretation-based
Monitoring	Satellite MRV workflow reduces field costs; adaptive validation recommended	Operational design
Recovery Window	3–5 year indicative trajectory — early intervention advised	Comparative benchmark
CSR Readiness	Pre-feasibility framework established — field validation pending	Feasibility phase

Pre-Feasibility Assessment. All indicators are satellite-derived observations requiring ground-truth validation. Quantitative values are based on actual GEE calculations for Jonnahalli Lake 2.

STUDY AREA & BASELINE ENVIRONMENTAL CONDITION

Geographic setting, ecological zone, and integrated environmental baseline.

Jonnahalli Lake 2 is situated at 13.1813° N, 77.7572° E within the Devanahalli region of Karnataka, encompassing 3,272.08 sq.m (0.327 ha / 0.81 acres). The site occupies a peri-urban catchment experiencing accelerating degradation driven by vegetation fragmentation, hydrological stress, thermal amplification, and urban encroachment along the Bengaluru–Devanahalli corridor.

ATTRIBUTE	DESCRIPTION
Location	Jonnahalli Lake 2, Devanahalli, Karnataka
Coordinates	13.1813° N, 77.7572° E
Ecological Zone	Semi-arid Deccan Plateau, peri-urban transition landscape
Project Area	3,272.08 sq.m 0.327 ha 0.81 acres
Climate	Tropical semi-arid with seasonal monsoon rainfall (Köppen: Aw)
Primary Stressors	Vegetation fragmentation, reduced water retention, sediment accumulation, urban encroachment
Ecological Role	Seasonal wetland, local groundwater recharge node, peri-urban biodiversity habitat



FIG 1.0 — STUDY AREA OVERVIEW — JONNAHALLI LAKE 2
True-color Sentinel-2 satellite composite. Green polygon = project boundary (3,272 sq.m). Site exhibits sparse vegetation, exposed surfaces, and surrounding peri-urban land-use. WGS84. May 2026.

NDVI BASELINE 0.238 Sparse vegetation	NDWI BASELINE −0.278 Moisture deficit	LST BASELINE 34.64°C Elevated thermal	PROJECT AREA 0.327 ha 3,272 sq.m 0.81 ac
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VEGETATION, LAND COVER & THERMAL ANALYSIS

Satellite-derived vegetation health, land cover classification, and thermal stress profiling.

NDVI analysis confirms sparse to moderately fragmented vegetation continuity (0.238 avg.). LULC classification identifies distinct intervention zones. LST profiling confirms elevated surface temperatures (34.64°C) consistent with limited vegetation cover and exposed soil conditions.



FIG 2.1 — NDVI VEGETATION HEALTH HEATMAP
Dark-green = dense vegetation (NDVI 0.45+). Yellow = sparse cover (0.15–0.30). Absence of dense-green signatures across the project boundary confirms fragmented biomass.

■ Dense 0.45+ ■ Mod 0.30–0.45 ■ Sparse 0.15–0.30 ■ Bare <0.15



FIG 2.2 — LAND SURFACE TEMPERATURE DISTRIBUTION
Uniformly elevated thermal load (avg. 34.64°C). Yellow-orange = high thermal stress consistent with bare ground. Cooler zones correspond to vegetated areas.

■ Hot >37°C ■ Warm 34–37°C ■ Moderate 30–34°C

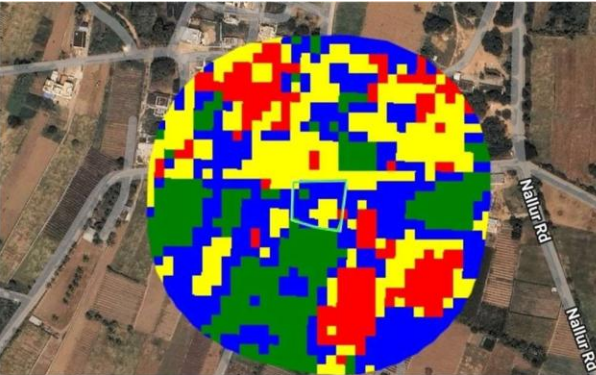


FIG 2.3 — LAND USE & LAND COVER CLASSIFICATION
Supervised LULC. Blue = water; Green = vegetation; Red = built-up; Yellow = bare soil. Fragmented pattern confirms ecological degradation across the catchment.

■ Water ■ Vegetation ■ Built-up ■ Bare soil



FIG 2.4 — NDWI SURFACE MOISTURE DISTRIBUTION
NDWI moisture mapping (avg. -0.278). Yellow-orange-red = moisture deficit. Cyan pixels indicate residual moisture near the lake margin. Orange-red zones require priority hydrological intervention.

■ Moist >0.1 ■ Low -0.1–0.1 ■ Deficit -0.3–0.1 ■ Stressed <-0.3

LAND COVER CLASS	DESCRIPTION	PRIORITY
Exposed Soil / Bare Ground	Disturbed open surfaces, cleared terrain, degraded buffer zones	HIGH
Sparse Vegetation	Scattered herbaceous / scrub; ecologically disconnected patches	MODERATE
Moderate Vegetation	Established shrub / grass; intact but ecologically isolated	LOW
Water Body	Seasonal / permanent water features; core asset requiring buffer protection	HIGH

OPERATIONAL INSIGHT

NDWI confirms significant moisture stress (avg. -0.278). Restoration must prioritise hydrological stabilisation — runoff interception and soil moisture improvement — before broad-front plantation is deployed.

HYDROLOGICAL & TERRAIN ASSESSMENT

Surface moisture distribution, runoff pathway analysis, and terrain morphology.

Surface moisture analysis reveals fragmented hydrological activity across the catchment. SRTM 30m DEM confirms a gently sloping surface (avg. slope 3.68°, elevation 880.59 m) that concentrates seasonal runoff along defined corridors converging toward the lake basin.

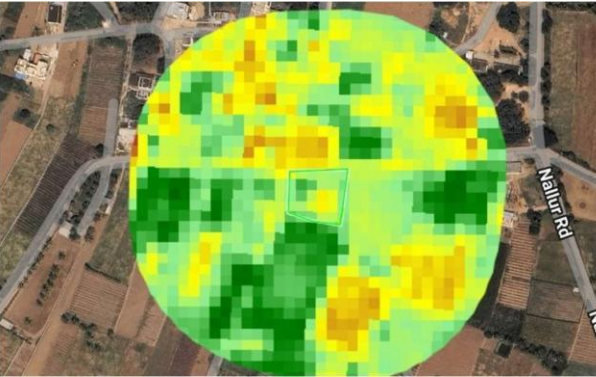


FIG 3.1 — SURFACE MOISTURE & WATER DISTRIBUTION

NDWI-based hydrological mapping. Green = localised moisture zones. Yellow-orange = hydrologically stressed. Red = highest moisture deficit — priority intervention zones = highest development pressure requiring boundary management and ecological buffer establishment.

Moisture present Low moisture Deficient Stressed

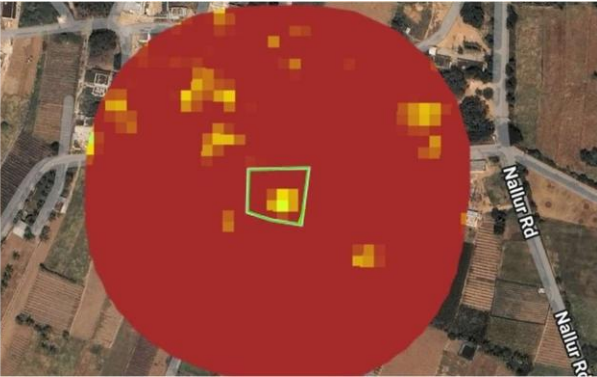


FIG 3.2 — URBAN ENCROACHMENT & BUILT-UP PRESSURE

Anthropogenic pressure mapping. Red = extensive built-up interaction. Yellow-orange = highest development pressure requiring boundary management and ecological buffer establishment.

High built-up Moderate Low pressure

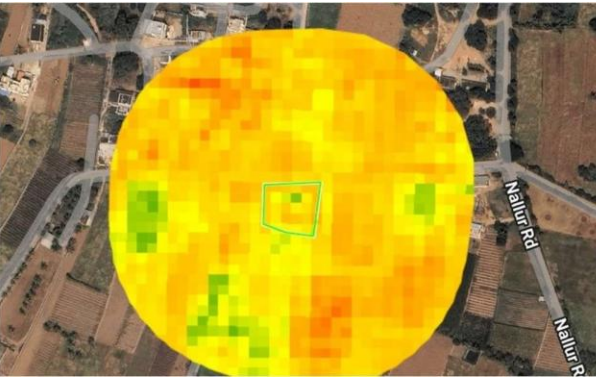


FIG 3.3 — RUNOFF ACCUMULATION & FLOW PATHWAYS

DEM-derived flow accumulation. Yellow-orange = runoff concentration corridors converging toward the lake. Key locations for earthen bunds and boulder check-dam interception infrastructure.

High accumulation Moderate Low / dispersed



FIG 3.4 — RESTORATION SUITABILITY INDEX

Composite suitability (score 0.389). Green = moderate-to-high restoration potential. Yellow = preparatory stabilisation required before active plantation. Catchment-wide feasibility confirmed.

High 0.5+ Mod. 0.35-0.5 Lower 0.2-0.35 <0.2

AVG ELEVATION 880.59 m Deccan Plateau terrain	AVG SLOPE 3.68° Low-to-moderate gradient	VEG. DENSITY 0.227 Below restoration threshold	MOISTURE DEFICIT -0.278 Active hydrological stress
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OPERATIONAL INSIGHT

The 3.68° slope creates well-defined drainage accumulation corridors identifiable through DEM. Runoff interception infrastructure at these locations delivers measurable hydrological benefit before vegetation canopy is established, protecting both the lake water body and nascent planting investments.

RESTORATION SUITABILITY & ECOLOGICAL STRESS

Composite spatial suitability assessment and multi-indicator stress profiling.

The composite suitability index (0.389) integrates NDVI, NDWI, LST, terrain stability, LULC, and urban pressure. The site-averaged score indicates low-to-moderate restoration feasibility — ecological recovery is achievable with appropriately sequenced intervention, pending preparatory stabilisation.

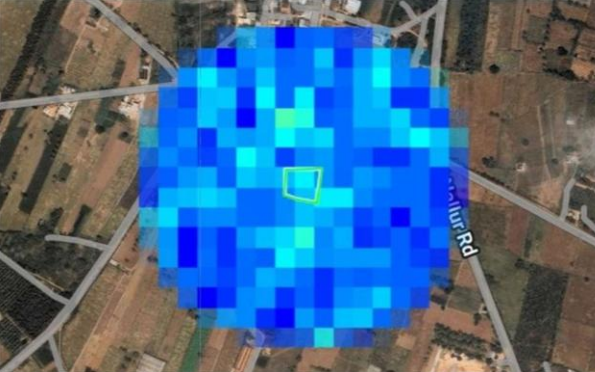


FIG 4.1 — RESTORATION SUITABILITY MAP
Composite suitability (0.389). Blue-cyan zones = restoration potential concentration. Lighter blue = lower-suitability zones requiring stabilisation prior to plantation.

High suitability

Moderate

Lower

Constrained

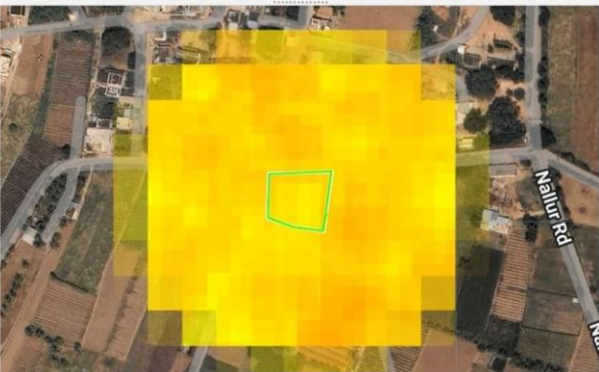


FIG 4.2 — ECOLOGICAL STRESS ASSESSMENT
Multi-indicator stress (score -0.240). Yellow-orange = active stress concentrated across the lake buffer and encroachment-adjacent transition zones.

High stress

Moderate-high

Moderate

Low stress

PRIORITY ZONE	CHARACTERISTICS	RECOMMENDED INTERVENTION	PHASE
HIGH PRIORITY	Degraded shoreline, bare-ground hotspots, active runoff corridors	Shoreline stabilisation, soil treatment, native ground cover, runoff interception	Phase 1
MODERATE	Partially vegetated peripheral zones, fragmented scrubland, moderate stress buffer	Native shrub and grass plantation following Phase 1 stabilisation	Phase 2
STABLE	Comparatively intact vegetation; active plantation not immediately required	Protective management, encroachment prevention, invasive species monitoring	Ongoing

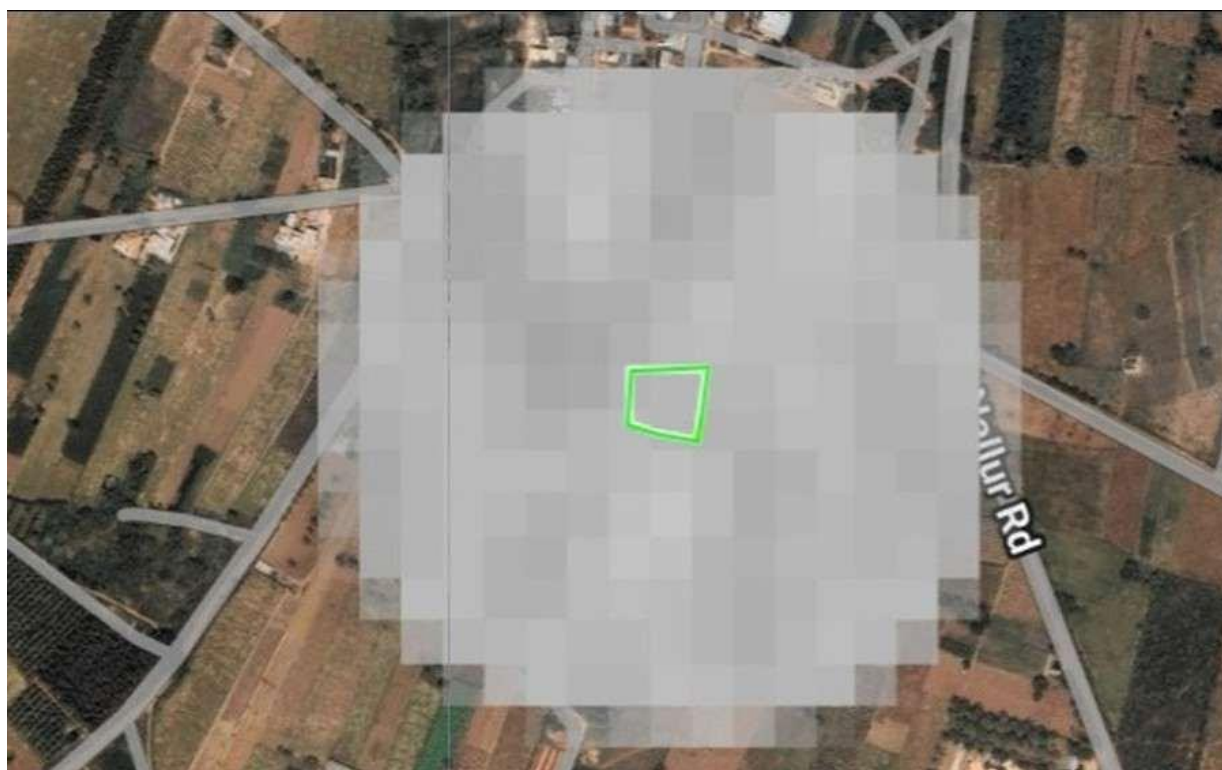


FIG 4.3 — RESTORATION INTERVENTION ZONING MAP

Spatial extent of restoration assessment area relative to the project boundary (green polygon) and surrounding catchment. Intervention zoning — high priority (shoreline / exposed), moderate (peripheral vegetation), stable (buffer) — is analytically defined from the composite assessment above. Spatial re-classification with field validation recommended before investment-grade design.

HISTORICAL ECOLOGICAL CHANGE ANALYSIS

Multi-year true-color satellite observation: 2019–2025.

Multi-temporal true-color imagery documents a sustained degradation trajectory at Jonnahalli Lake 2 across a five-year observational window. The record confirms progressive hydrological variability, vegetation loss, and urban encroachment — establishing that active intervention is required rather than passive natural recovery.



FIG 5.0 — MULTI-TEMPORAL ECOLOGICAL CHANGE — JONNAHALLI LAKE 2 (2021–2025)

Six true-color satellite observations document progressive change from comparatively intact 2021 conditions toward the current degraded baseline. Observable transitions include shoreline recession, vegetation loss, increasing exposed sediment, and advancing urban boundary. Yellow polygon = project monitoring boundary.

YEAR	CONDITION	KEY OBSERVATION
2019 — Baseline	More intact vegetation; lake margins with moderate riparian cover	Reference condition — vegetation connectivity partially intact, urban boundary set back
2020	Early fragmentation; urban development advancing	Vegetation patch connectivity reducing; bare-ground exposure emerging
2021	Accelerating vegetation loss; NDVI declining; fragmentation intensifying	Connected patches becoming spatially isolated; thermal signatures increasing
2022	Consolidated degradation; sparse patches across exposed landscape	Urban encroachment advances; impervious surface runoff loading increases
2023	Sustained pressure; no meaningful natural recovery	Absence of passive recovery confirms active intervention requirement
2025 — Current	NDVI 0.238, NDWI -0.278, LST 34.64°C	Critical intervention window: degradation trajectory continues without restoration

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QUANTIFIED ECOLOGICAL METRICS & BENCHMARK DASHBOARD

Satellite-derived site-scale benchmarking and recovery indicator framework.

Benchmark comparison provides context for the restoration gap. Current conditions fall between degraded and healthy reference states across all primary metrics, confirming moderate-to-achievable restoration potential through structured phased intervention.

ECOLOGICAL METRIC	CURRENT	DEGRADED	HEALTHY	5-YR TARGET
NDVI	0.238	0.15–0.25	0.50–0.70	0.40–0.52
NDWI	–0.278	–0.40—0.25	0.10–0.40	+0.05 to +0.18
LST (°C)	34.64	34–37	26–30	30–32.5
Vegetation Cover	~23%	10–20%	60–80%	38–52%
Restoration Suitability	0.389	0.15–0.30	0.60–0.85	0.50–0.65
Ecological Stress	–0.240	–0.35—0.20	< –0.05	–0.10—0.05
Vegetation Density	0.227	< 0.20	> 0.55	0.38–0.48

CURRENT STATE → 5-YEAR RECOVERY TARGETS

NDVI 0.238 Sparse vegetation → 0.40–0.52 Moderate recovery	NDWI –0.278 Limited moisture → +0.05–+0.18 Improved persistence	LST 34.64°C Elevated thermal → 30–32.5°C Reduced stress	ECO. STRESS –0.240 Moderate stress level → –0.10—0.05 Stabilised
5-year targets represent realistic recovery milestones based on regional restoration case studies. Progress monitored through the MRV framework with quarterly satellite assessments and annual field validation.			

NEVARA ANALYTICAL METHODOLOGY

COMPONENT	METHODOLOGY	SOURCE	OUTPUT
NDVI	(NIR–Red)/(NIR+Red) ratio	Sentinel-2 Band 8/4	Vegetation density map
NDWI	(Green–NIR)/(Green+NIR)	Sentinel-2 Band 3/8	Surface moisture map
LST	Thermal infrared; emissivity correction	Landsat 8/9 TIRS	Temperature distribution
LULC	Supervised multi-spectral classification	Sentinel-2 10m	Land cover categories
DEM / Terrain	Slope, hillshade, flow accumulation	NASA SRTM 30m	Runoff pathway maps
Suitability Index	NDVI + NDWI + LST + terrain + LULC composite	All layers	Priority zoning

OPERATIONAL RESTORATION INTELLIGENCE

Phased intervention strategy, restoration masterplan, and cost-to-impact analysis.

The site profile — sparse vegetation, negative moisture index, moderate thermal stress, fragmented land cover — indicates a landscape where restoration resources deployed without spatial prioritisation produce inconsistent outcomes. Targeted, preparatory intervention is operationally optimal.

INTERVENTION TYPE	COST	EXPECTED ECOLOGICAL IMPACT	PRIORITY
Shoreline Stabilisation (vetiver + native sedge)	Low–Mod	HIGH — immediate erosion arrest, shoreline protection, establishment conditions	HIGHEST
Runoff Interception (bunds / boulder checks)	Low–Mod	HIGH — reduces erosional velocity, promotes infiltration and moisture retention	HIGH
Native Grass Plantation	Low	MODERATE–HIGH — rapid ground cover, soil binding, thermal stress reduction	HIGH
Invasive Vegetation Removal	Very Low	MODERATE — reduces competitive suppression; triggers natural regeneration	HIGH
Peripheral Revegetation (native shrub)	Moderate	HIGH (medium-term) — restores vegetation continuity, reduces fragmentation	MEDIUM
Thermal Buffer Planting	Moderate	MODERATE — shade-creating native species progressively reduce LST	MEDIUM
Sediment Control Measures	Low–Mod	MODERATE — reduces turbidity and sediment accumulation in lake	MEDIUM

PHASE	TIMELINE	PRIMARY ACTIVITIES	DEPENDENCIES
Phase 1 — Stabilisation	Months 1–6	Shoreline bio-engineering; runoff interception; native grass; invasive removal; mulching	Field validation
Phase 2 — Hydrology	Months 6–18	Native shrub plantation; thermal buffer; sediment control; NDVI/NDWI monitoring cycle	Phase 1 complete
Phase 3 — Recovery	Years 2–5	Native tree canopy; aquatic fringe vegetation; biodiversity features; annual audit	Phase 2 established
Phase 4 — Self-Sustaining	Years 5+	Self-sustaining ecosystem; community stewardship; MRV-based verification reporting	All prior phases

ZONE	SPATIAL LOCATION	PRIMARY ACTIVITIES	OBJECTIVE
A — Shoreline	Lake margin / immediate shoreline	Vetiver, native sedge, jute geotextile, boulder toe protection	Arrest erosion; stable substrate
B — Runoff Intercept	DEM flow accumulation corridors	Earthen bunds, boulder checks at 20–30m intervals, infiltration trenches	Reduce runoff velocity; infiltration
C — Native Plantation	Peripheral buffer 20–50m	Native grass mix, progressive shrub canopy, mulching	Ground cover, thermal reduction
D — Thermal Buffer	Exposed bare-ground zones	Neem, Pongamia, native Ficus canopy establishment	LST reduction; microclimate
E — Biodiversity	Northern boundary connectivity strip	Native shrub mix, habitat features, nesting provisions	Ecological connectivity
F — Invasive Control	All disturbed surfaces	Manual Lantana / Parthenium removal; competitive native planting immediately following	Enable native regeneration

MONITORING FRAMEWORK, RISKS & CSR READINESS

MRV architecture, satellite monitoring schedule, risk register, and CSR alignment.

PARAMETER	FREQUENCY	SATELLITE SOURCE	MONITORING PURPOSE
NDVI — Vegetation Health	Monthly	Sentinel-2 MSI (10m)	Recovery tracking; plantation survival
NDWI / Moisture	Seasonal	Sentinel-2 MSI (10m)	Moisture retention trends; hydrological regime
LST — Thermal Condition	Quarterly	Landsat 8/9 TIRS (30m)	Thermal stress; canopy cooling verification
LULC — Land Cover	Annual	Sentinel-2 MSI (10m)	Change detection; encroachment alert

RISK CATEGORY	DESCRIPTION	LIKELIHOOD	IMPACT	MITIGATION
Seasonal Variability	Sub-average monsoon reduces establishment survival	MOD	HIGH	Drought-tolerant species; dry-season watering Year 1
Urban Encroachment	Settlement boundary threatens buffer integrity	HIGH	HIGH	Boundary demarcation; community engagement Phase 1
Invasive Recolonisation	Lantana / Parthenium recolonise cleared zones	HIGH	MOD	Annual monitoring; competitive native planting immediately after removal
Hydrological Uncertainty	Groundwater depth unconfirmed; affects projections	MOD	MOD	Hydrological survey recommended (2–4 months)
Maintenance Dependency	Gains reverse without consistent Year 1–3 maintenance	MOD	HIGH	Community stewardship; MRV-triggered maintenance alerts
Field Validation Gap	No ground-truth survey conducted; satellite only	N/A	MOD	Field validation survey (3–6 months) before detailed design

READINESS FACTOR	CURRENT STATUS	REQUIRED FOR INVESTMENT
Satellite Baseline	Complete — NDVI, NDWI, LST, Suitability verified	Available now
Ecological Intelligence	Pre-feasibility assessment complete	Available now
Field Validation	Pending — recommended before investment commitment	3–6 months
Hydrological Survey	Pending — required for intervention design	2–4 months
Community Engagement	Not initiated	3–6 months
Detailed Cost Estimation	Indicative — requires survey inputs	Following field validation

5-YEAR OUTLOOK, ESG ALIGNMENT & CONCLUSION

Projected recovery trajectory, ESG contributions, executive conclusion, and document control.

YEAR	NDVI TARGET	NDWI TARGET	LST TARGET	MILESTONE
Year 1	0.25–0.30	+0.00–0.20	33–34.5°C	Ground cover established; shoreline stabilised; bare-ground hotspots treated
Year 2	0.30–0.36	–0.10+0.05	32–33°C	Shrub advancing; moisture retention improving in treated zones
Year 3	0.34–0.40	+0.02+0.10	31–32.5°C	Vegetation continuity partially restored; thermal buffer measurable
Year 4	0.38–0.45	+0.05+0.15	30–32°C	Near-canopy closure; biodiversity indicators improving
Year 5 (Target)	0.40–0.52	+0.05+0.18	30–32.5°C	Moderate veg. recovery; improved moisture persistence; reduced thermal stress

SDG ALIGNMENT & ESG CONTRIBUTION

6	11	13	15
CLEAN WATER	SUSTAINABLE CITIES	CLIMATE ACTION	LIFE ON LAND

ESG CONTRIBUTION	ENVIRONMENTAL VALUE	REPORTING RELEVANCE
Biodiversity Improvement	Enhanced habitat connectivity and native species establishment	ESG biodiversity metrics
Urban Cooling Potential	Reduced peri-urban heat island through canopy establishment	Climate adaptation reporting
Groundwater Recharge	Improved aquifer replenishment through catchment moisture recovery	Water stewardship disclosure
Native Plantation	Indigenous vegetation communities; low-maintenance ecological substrate	Land use & forestry reporting
Ecological Resilience	Adaptive capacity enhancement in peri-urban transition landscape	Climate risk assessment

RESTORATION SUITABILITY	RECOVERY TIMELINE	INTERVENTION MODE	ESG READINESS
0.389	3–5 yrs	Moderate	Pre-feasibility
Class 2 — Moderate potential	Observable improvement horizon	Focused priority zone action	Field validation pending

CONCLUSION

This Assessment establishes a comprehensive satellite-derived baseline for Jonnahalli Lake 2 — a lake ecosystem under moderate environmental stress with viable restoration potential. NDVI 0.238, NDWI –0.278, and LST 34.64°C collectively indicate a landscape on a reversible degradation trajectory. Priority interventions: shoreline stabilisation, runoff interception, and native ground cover in Phase 1. Meaningful ecological recovery is achievable over a 3–5 year programme horizon. Field validation and hydrological survey are recommended before investment commitment.

OPERATIONAL CONCLUSION
"Jonnahalli Lake 2 demonstrates recoverable ecological functionality. Verified satellite observations (NDVI 0.238, NDWI –0.278, LST 34.64°C) confirm the site is suitable for structured restoration intervention. The NEVARA MRV framework provides the operational intelligence infrastructure for planning, monitoring, and verification. Field validation is recommended as the next step before investment commitment."

DOCUMENT CONTROL

PARAMETER	DETAIL
Reference	NEV-RIA-JHL2-2026-002
Version	1.0 — Pre-Feasibility Assessment
Date	May 2026
Location	Jonnahalli Lake 2, Devanahalli, Karnataka
Classification	Confidential Environmental Intelligence Document
Status	Pre-Feasibility — Field validation pending

PARAMETER	DETAIL
Next Review	Following hydrological survey (recommended within 6 months)

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